

**Benthic Substrate-Associated eDNA Metabarcoding Reveals Aquatic Insect
Community Responses to Water Quality Gradients in the Nakdong River
Basin, South Korea**

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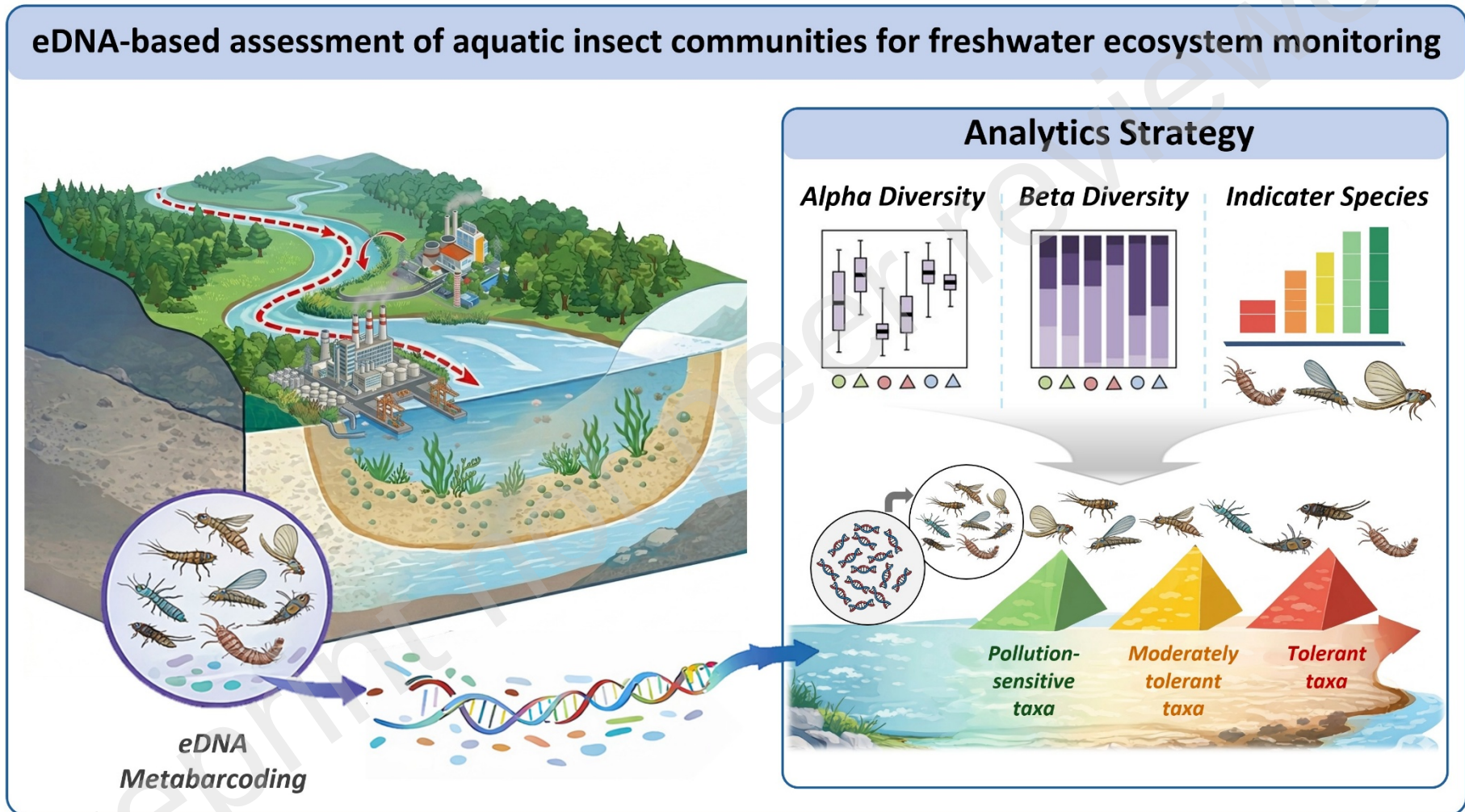
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Abstract

Biomonitoring using aquatic insects is a key component of freshwater ecosystem management. While environmental DNA metabarcoding has emerged as a high-resolution tool, its practical application for developing quantitative water quality indicators necessitates further validation. This study evaluated the efficacy of substrate-associated environmental DNA-based assessments of aquatic insects in representing physicochemical water quality gradients across the Nakdong River basin, South Korea. Using mitochondrial cytochrome oxidase subunit I metabarcoding, we analyzed 48 high-quality samples from 30 sites, validated through rigorous quality filtering. Community responses to environmental gradients were assessed through three analytical strategies: alpha diversity dynamics, beta diversity structures comparing phylogenetic and taxonomic metrics, and indicator species analysis. Our results revealed distinct community responses to varying levels of organic and nutrient loading. Specifically, species richness significantly declined as total phosphorus, total organic carbon, and ammonia-nitrogen increased. Beta diversity analysis demonstrated that abundance-weighted phylogenetic diversity metrics provided superior explanatory power, significantly outperforming traditional taxonomic dissimilarity in discriminating degraded sites. Taxonomic composition shifted significantly along the water quality gradient, transitioning from balanced communities in pristine sites to a high relative abundance of Diptera in degraded areas. Furthermore, indicator species analysis identified several key genera that effectively represent different water quality groups, including both pollution-sensitive and tolerant taxa. Our findings demonstrate that environmental DNA-based assessment accurately reflects the physicochemical status of riverine ecosystems, providing a reliable, data-driven framework for the next-generation monitoring and management of freshwater biodiversity.

46 **Keywords:** Environmental DNA metabarcoding; Aquatic insect; Phylogenetic community
47 structure; Indicator species; Water quality assessment



1. Introduction

The overexploitation of natural resources and extensive land-use changes by humans have profoundly impacted global ecosystems, resulting in the degradation of habitat quality and a significant loss of biodiversity (Dasgupta, 2021; Krausmann et al., 2013). These alterations trigger cascading effects that exacerbate climate change and accelerate species extinction, ultimately disrupting ecological communities and impairing vital ecosystem services, including the provision of drinking water, irrigation, and food resources (Ceballos and Ehrlich, 2023; Ceballos et al., 2017; Deeksha and Shukla, 2022; Díaz et al., 2019). Despite the urgency, rapidly predicting and monitoring community responses to ongoing environmental changes remains a significant challenge, largely due to the inherent limitations of traditional assessment tools (Bitter et al., 2021; Thompson et al., 2021).

For over a century, biodiversity monitoring has served as the cornerstone for assessing the health of aquatic ecosystems (Rosenberg and Resh, 1992). To align with the stringent demands of the 2021 Global Agenda on biodiversity monitoring, there is a growing paradigm shift toward innovative approaches, such as omics-based technologies, artificial intelligence, and advanced bioinformatics (Maasri et al., 2022). It is now essential to move beyond simple data collection and establish integrative and quantifiable biodiversity indicators that can explicitly link ecosystem responses to multiple anthropogenic stressors (Birk et al., 2020; Díaz et al., 2020). Environmental DNA (eDNA) metabarcoding has recently emerged as an effective solution for biomonitoring and ecosystem assessments, particularly for aquatic systems (Fernández et al., 2019; Lee et al., 2024; Pikitch, 2018; Uchida et al., 2020). eDNA can provide a relatively high sensitivity and the ability to identify cryptic and rare species that are often missed by conventional surveys (Çevik and Çevik, 2025; David et al., 2021). Although eDNA has been validated as a potent monitoring tool, the establishment of robust ecosystem health

indicators derived from eDNA-based datasets remains still in its infancy (Cook et al., 2025; Wilkinson et al., 2024).

In South Korea, the Ministry of Climate, Energy and Environment (MCEE) operates an extensive nationwide monitoring infrastructure, maintaining a network of 1,953 water quality stations—including 691 river and 201 lake sites—alongside 3,035 annual aquatic ecosystem health assessment sites. However, the two monitoring programs—the Aquatic Ecosystem Health Assessment and National Ecosystem Survey—are often managed in isolation. Consequently, a practical framework to integrate physicochemical variables with biological community responses has yet to be established (MCEE/NIBR, 2016-2025). Most assessments treat environmental and biological data as independent datasets, lacking a quantitative method to link stressors with ecological responses.

As a dominant taxonomic group of benthic macroinvertebrates, aquatic insects serve as ideal bioindicators for addressing this gap. Their limited mobility ensures that they reflect localized environmental conditions/stressors, while their high sensitivity to disturbances provides an accurate portrayal of the integrated ecological status of their habitats (Rodríguez et al., 2024; Uchida et al., 2020). While these communities have long been utilized in freshwater biomonitoring, conventional surveys—dependent on specimen collection and morphological identification—are often hindered by labor-intensive sampling effort, lengthy processing periods, coupled with the critical requirement for specialized taxonomic expertise (Emilsson et al., 2017). To address these limitations, eDNA metabarcoding has been increasingly applied to detect aquatic insects. In particular, metabarcoding of unsorted, unpicked kick-net bulk samples (substrate-associated eDNA) can recover substantially higher macroinvertebrate diversity than either sorted specimens or water-column eDNA, likely because many aquatic insects are closely associated with benthic substrates (Pereira-da-Conceicao et al., 2021).

Therefore, substrate-associated eDNA metabarcoding offers a powerful solution to these challenges by enabling the efficient and standardized characterization of aquatic insect assemblages across broad environmental gradients. This approach effectively bridges the gap between physicochemical water quality data and biological health assessments, providing an integrated framework for river management (Uchida et al., 2020).

In this study, we aimed to evaluate whether substrate-associated eDNA-based biodiversity assessments of aquatic insect communities can provide a quantitative framework for linking physicochemical water quality parameters with community-level responses. The specific objectives were to: 1) characterize alpha diversity dynamics across distinct water quality groups and identify key environmental drivers of community richness; (2) compare the efficacy of community-level phylogenetic metrics (Weighted Unifrac distances) versus traditional taxonomic metrics (Bray-Curtis distances) in capturing water quality gradients; (3) identify pollution-sensitive and tolerant indicator taxa to validate the diagnostic potential of eDNA for next-generation biomonitoring frameworks.

2. Materials and Methods

2.1 Study area

The Nakdong River, the longest river in South Korea (approximately 510 km), serves as a vital water source for the southeastern region, supporting the domestic, industrial, and agricultural needs of more than 10 million people (Figure 1). Its mid-to-downstream reaches, which encompass major metropolitan and industrial hubs, including Busan, Daegu, Gimhae, Pohang, and Ulsan, are characterized by high population density and extensive industrial complexes (e.g., electronics, textiles, and machinery). Consequently, these areas necessitate rigorous monitoring to mitigate the potential impacts of industrial and municipal wastewater on the aquatic ecosystem (Jung et al., 2016).

A total of 30 monitoring sites were selected to evaluate benthic macroinvertebrate communities (Figure 1; Table 1). Twenty of these sites were distributed across the Nakdong River basin categorized by reach (upper, middle, and lower), water quality grades, and environmental characteristics (natural, industrial, and urban). To further investigate community responses to localized water quality fluctuations and diverse land-use patterns, an additional 10 sites were established in key tributary systems within the Gimhae region, an area characterized by intense anthropogenic pressure (Figure 1). These sites encompass major tributary streams and their confluences, providing a comprehensive gradient of environmental impact.

2.2 Field sampling

Field surveys were conducted twice, in May and September 2025. At each site, 4L of surface water was collected for physicochemical water quality assessment. On-site environmental

parameters, including water temperature, dissolved oxygen (DO), pH, and specific conductivity (SPC) were measured directly using ProDSS Multiparameter Digital Water Quality Meter (YSI Inc., USA). For further chemical analysis, the collected water samples were transported to a specialized analytical agency. The concentrations of biological oxygen demand (BOD), total nitrogen (TN), total phosphorus (TP), ammonia-nitrogen ($\text{NH}_3\text{-N}$), nitrate-nitrogen ($\text{NO}_3\text{-N}$), phosphate-phosphorus ($\text{PO}_4\text{-P}$), and total organic carbon (TOC) were analyzed following standard methods.

For eDNA metabarcoding analysis, bulk benthic macroinvertebrate samples were collected using standardized sampling methods according to the water depth: a 500 μm kick net was utilized for sites with water depths exceeding 30cm, while a 500 μm surber net was employed for shallower areas under 30cm. The methods target eDNA settled on the riverbed. Substrates such as gravel, sand, sediment, and detritus are collected in a surber net, and the eDNA attached the gravel surfaces or within the sediment is subsequently isolated by separating the mixture manually (NNIBR, 2024). The accumulated organic debris samples from the surber net were placed into plastic wide-mouth bottles and preserved in 70% ethanol. To prevent cross-contamination between samples, new gloves and sterilized surber nets were used at each site. In cases where equipment was reused, tools were thoroughly rinsed with site water to eliminate DNA residues from previous sites.

All sampling procedures, including eDNA-related protocols, were performed in accordance with “A practical Guide to Environmental DNA-based Survey and Analysis of Aquatic Insect Diversity” published by the Nakdonggang National Institute of Biological Resources (NNIBR, 2024).

2.3 DNA extraction and PCR amplification

Benthic bulk samples were first processed by removing large debris, such as gravel and stones. Macroorganisms visible to the naked eye were intentionally removed to minimize biomass-driven bias in the metabarcoding results. The remaining samples were then homogenized using a blender at maximum RPM for 20–30s. For each sample, three 1 mL aliquots of the homogenate were transferred to microcentrifuge tubes and centrifuged at $15,000 \times g$ for 1 minute to remove the ethanol supernatant, followed by drying at 70°C to ensure complete evaporation. Work surfaces and equipment were decontaminated with 10% sodium hypochlorite followed by rinsing with distilled water between samples to prevent cross-contamination. As procedural negative controls, sterilized distilled water ('blank') was subjected to the same processing steps before and after sample processing (spring: $n = 2$; autumn: $n = 2$) and extracted alongside the field samples. DNA was extracted using the DNeasy PowerSoil Pro Kit (Qiagen, Germany) following the manufacturer's protocol with specific modifications. For each site, three replicate PowerBead Pro Tubes were prepared by adding the dried homogenate with 800 μL of CD1 solution. To maximize the final DNA concentration, the lysates from the three replicates were pooled into a single MB Spin Column following the addition of CD3 solution. The eDNA was then recovered through two successive elutions using 50 μL of C6 solution (total 100 μL), with a 10-minute incubation period incorporated into each solution step to enhance recovery. The concentration of the extracted DNA was determined using a NanoDrop One (Thermo Fisher Scientific, USA). The mean DNA concentrations were 55.4 ng/ μL for the field samples.

For DNA metabarcoding, the mitochondrial *COI* region was amplified using the mlCOIintF (5'-GGWACWGGWTGAACWGTWTAYCCYCC-3') and jgHCO2198 (5'-TAIACYTCIGGRTGICCAARAAYCA-3') (Geller et al., 2013; Leray et al., 2013). PCR amplification was performed under the following thermal cycling conditions: an initial denaturation at 95°C for 5 min; followed by 34 cycles of denaturation at 95°C for 30 s,

annealing at 43°C for 30 s, and extension at 72°C for 50 s; with a final extension step at 72°C for 5 min. PCR products were visualized by gel electrophoresis to confirm successful amplification at the target amplicon size for the field samples, while no visible bands were detected in the negative controls (or procedural blanks). The resulting PCR products were purified using the QIAquick PCR Purification Kit (Qiagen, Germany) to remove residual reagents and primers. Finally, paired-end sequencing (2 x 300 bp) was conducted on the Illumina MiSeq platform at Macrogen Inc. (Seoul, Korea).

2.4 Bioinformatics and Statistical Analysis

Metabarcoding data were processed using the QIIME2 pipeline (Bolyen et al., 2019). Raw sequence data were first treated with Cutadapt (Martin, 2011) to identify and remove primer sequences. The resulting primer-free reads were then subjected to the DADA2 plugin (Callahan et al., 2016) for Amplicon Sequence Variant (ASV) inferences, which included quality filtering, denoising, and chimera removal. Taxonomic assignment of the ASVs was performed using BLAST+ (Camacho et al., 2009) by querying the sequences against a comprehensive reference database. This database integrated *COI* sequences of Korean aquatic insects retrieved from NCBI GenBank and BOLD, further supplemented by a specially curated in-house reference library of South Korean aquatic insect *COI* sequences to enhance the precision of local species identification. The in-house database includes Ephemeroptera, Plecoptera, Trichoptera, Diptera, Odonata, Coleoptera, and Hemiptera (NNIBR, 2024). For downstream analysis, a phylogenetic tree was constructed using the q2-phylogeny plugin. The ASVs were aligned with MAFFT and filtered to remove highly variable positions, followed by tree inference using FastTree. Based on this phylogeny, core diversity metrics were calculated using the q2-diversity plugin at a standardized sampling depth. Within the QIIME2 pipeline, alpha diversity indices—including

Observed features, Shannon index, Faith's phylogenetic diversity, and Pielou's evenness—
were calculated to analyze and compare community-level diversities across samples.

To perform comparative statistical analyses, water quality groups were assigned to
each site based on the "Environmental Quality Standards for River Water" provided by the
Water Resources Management Information System (WRMIS, Korea). Under the Korean
national river water quality standard, sites are classified into seven levels—Very Good, Good,
Slightly Good, Normal, Slightly Bad, Bad, and Very Bad—based on threshold values for key
physicochemical variables. Each level represents progressively degraded ecological integrity
and restricted water-use suitability (WRMIS, Korea). Following the national evaluation
protocol, the final water quality grade for each site was determined by the most degraded
(lowest) level among the measured physicochemical parameters, ensuring a conservative
assessment of environmental health. To ensure suitable sample size per group, enhance
statistical power, and facilitate a clearer ecological interpretation of biodiversity patterns along
the water quality gradient, the seven initial levels were consolidated into three broader
categories: Group 'A' (Very Good and Good), Group 'B' (Slightly Good, Normal, and Slightly
Bad), and Group 'C' (Bad and Very Bad).

All subsequent statistical analyses and data visualizations were performed in R
(v4.2.2). The ASV table, taxonomic information, and rooted phylogenetic tree were integrated
into a phyloseq package (McMurdie and Holmes, 2013). Alpha diversity indices, initially
calculated in QIIME 2, were visualized using ggplot2, and their correlations with
environmental variables (e.g., BOD, TN, TP, TOC) were evaluated using Spearman's rank
correlation via the Hmisc package (Harrell Jr and Harrell Jr, 2019). To further investigate the
drivers of community depletion, regression analyses were performed between richness and key
water quality parameters. Statistical significance of alpha diversity across the three water
quality groups was determined using the Kruskal–Wallis test, followed by Dunn's test as a

post-hoc procedure for pairwise comparisons. To investigate among-group community structure, beta diversity was assessed using Weighted UniFrac distances and Bray-Curtis dissimilarity. Significant differences in community composition among water quality groups were tested using PERMANOVA (adonis2) and PERMDISP (betadisper) via the vegan package (Oksanen et al., 2013). Constrained Analysis of Principal Coordinates (CAP) was performed to visualize the influence of environmental factors on community assembly, with significant environmental vectors identified through the envfit function. Furthermore, taxonomic composition was visualized at the order level, and the relationship between dominant genera and environmental variables was analyzed using Spearman correlation heatmaps. Finally, indicator species analysis was conducted using the indicpecies package (De Cáceres et al., 2016) to identify taxa significantly associated with specific water quality groups.

3. Results

3.1. Sequencing Overview and Dataset Validation

A total of 8,647,935 raw reads were generated via *COI* metabarcoding, which were subsequently refined to 7,273,139 high-quality reads after undergoing stringent bioinformatics processing, including quality filtering and chimera removal. The number of reads per sample ranged from 37,393 to 229,846, with a mean of 121,219. To ensure that the community analysis specifically reflected aquatic insect diversity, the dataset was filtered to retain only those sequences assigned to nine aquatic insect orders (Ephemeroptera, Plecoptera, Trichoptera, Diptera, Odonata, Coleoptera, Hemiptera, Lepidoptera, and Megaloptera), with read counts per

sample ranging from 45 to 40,584, with a mean of 11,794. To improve statistical accuracy, twelve samples with insufficient sequencing depths ($< 3,000$ read counts) for the aquatic insects were excluded through rarefaction-based normalization, leaving a robust final dataset of 48 samples for downstream ecological assessment (Table 1, Supporting Information Figure S1).

3.2. Physicochemical Characterization of Water Quality Groups

The physicochemical parameters measured across the 48 samples showed distinct variations according to the assigned water quality group (Table 2, Table S1). Overall, concentrations of organic matter and nutrients increased progressively from Group A to Group C. Notably, for most parameters—with the exception of DO—the ranges of variation showed no overlap among the groups, suggesting that each water quality group is characterized by a distinct environmental profile.

The concentration of organic pollutants differed significantly among Group A, Group B and Group C. BOD levels in Group A were the lowest (0.867 ± 0.166 mg/L), indicating pristine conditions, while Group C showed approximately ten times higher concentrations (9.18 ± 1.508 mg/L), reflecting severe organic loadings. TOC concentration followed a similar trend, with Group C (10.36 ± 1.541 mg/L) exhibiting nearly five times higher in organic carbon content compared to Group A (1.85 ± 0.191 mg/L). DO levels remained relatively high across all groups but were the lowest in Group C (8.139 ± 0.935), due to the oxygen consumption required for the decomposition of high organic matter.

Nutrient concentrations also demonstrated a marked increase in more degraded water quality groups. TN levels rose from 0.911 ± 0.171 mg/L in Group A to 5.010 ± 1.699 mg/L in Group C. Notably, $\text{NH}_3\text{-N}$ in Group C (2.355 ± 1.381 mg/L) showed significantly higher mean values with larger variation compared to Group A (0.033 ± 0.003 mg/L), suggesting localized inputs of nitrogenous waste in poor-quality sites. TP, a key indicator of eutrophication, showed

a more than ten-fold increase from Group A (0.018 ± 0.002 mg/L) to Group C (0.208 ± 0.034 mg/L). $\text{PO}_4\text{-P}$ was exclusively detected in Group C (0.04 ± 0.020 mg/L), further confirming the high nutrient pressure in these areas.

3.3 Alpha Diversity Dynamics

The alpha diversity indices of aquatic insect communities—including Ephemeroptera, Plecoptera, Trichoptera, Diptera, Odonata, Coleoptera, Hemiptera, Lepidoptera, and Megaloptera—were compared across the three water quality groups to assess the impact of water quality on species richness and evenness (Figure 2A). The richness of aquatic insects showed significant differences among the water quality groups ($p < 0.05$). Specifically, Group A exhibited a significantly higher richness compared to Groups B and C. In contrast to Richness, other diversity indices (e.g., Phylogenetic Diversity, Shannon index, and Evenness) did not show significant differences among the groups. In particular, Richness values showed a significant negative correlation with TP ($r = -0.41, p < 0.01$), TOC ($r = -0.35, p < 0.05$), and $\log \text{NH}_3\text{-N}$ ($r = -0.34, p < 0.05$) (Figure 2B).

3.4 Beta Diversity and Taxonomic Characteristics

The CAP analysis based on Weighted UniFrac distances indicated variation in aquatic insect community structure among the water quality groups (Figure 3A). The abundance-weighted phylogenetic structure of aquatic insect communities was significantly influenced by environmental variables ($R^2 = 56.3\%, p < 0.01$; Figure 3A). Group A sites clustered together, showing a negative association with organic and nutrient loads, while Group C samples exhibited high dispersion and a strong positive correlation with BOD, $\text{NH}_3\text{-N}$, TN, TOC, and TP. Pairwise PERMANOVA further confirmed significant differences in community distances

across all water quality groups ($p < 0.05$; Table S2 in the supplement). PERMDISP analysis showed no significant differences in multivariate dispersion among water quality groups for Weighted UniFrac distances ($F = 0.166$, $p = 0.851$), supporting the interpretation of the PERMANOVA results as differences in community structure among water quality groups. In contrast, CAP analysis based on Bray-Curtis distances showed more limited responses, with significant correlations observed only for BOD, TOC, and TP ($R^2 = 36.6\%$, $p < 0.01$; Figure S2). While pairwise PERMANOVA based on Bray-Curtis distances revealed significant differences between Group A and Groups B and C ($p < 0.05$), no significant difference was found between Group B and Group C ($p > 0.05$; Table S2). PERMDISP detected significant differences in multivariate dispersion among water quality groups for Bray-Curtis dissimilarities ($F = 6.701$, $p = 0.003$), indicating that the Bray-Curtis PERMANOVA results should be interpreted with caution because differences in within-group dispersion may have contributed to the observed group differences.

Taxonomic composition at the order level also varied across the water quality groups (Figure 3B). In Group A, Diptera (44.9%) and Ephemeroptera (43.0%) were the dominant orders, together accounting for nearly 88% of the whole community. In Group B, relative abundance of Ephemeroptera decreased to 27.8%, while Diptera increased to 53.1%. Notably, groups of Coleoptera (6.3%), Megaloptera (4.7%), and Odonata (4.7%) showed higher proportions in Group B compared to Group A. In Group C, Diptera became the overwhelmingly dominant order, representing 81.6% of the total abundance. While sensitive taxa such as Plecoptera and Megaloptera completely disappeared, Odonata (7.9%) and Ephemeroptera (6.6%) remained as minor components. Trichoptera, which accounted for 7.0% in Group A, was reduced to less than 1% (0.86%) in Group C, further confirming the shift toward pollution-tolerant taxa in degraded water quality.

3.5 Correlation between Environmental Variables and Key Genera

To further elucidate the relationship between environmental pressure and specific taxonomic groups, a Spearman correlation heatmap was constructed for the top 30 abundant genera (Figure 4). The analysis revealed two distinct clusters of genera based on their response to nutrient and organic loadings.

A distinct cluster of genera, belonging to specific members of Ephemeroptera and Diptera, exhibited significant negative correlations with TP, BOD, $\text{NH}_3\text{-N}$, and TOC (Figure 4). This correlation pattern was further supported by the indicator species analysis, which identified these genera as significant indicator taxa for higher water quality groups (Table S3). For instance, genera such as *Ecdyonurus* (Indicator Value = 0.762, $p = 0.001$) and *Rheopelopia* (Indicator Value = 0.789, $p = 0.001$) were identified as significant indicator taxa for Group A (Figure 4, highlighted in blue). Furthermore, *Ephemera* (Indicator Value = 0.753, $p = 0.025$) and *Teloganopsis* (Indicator Value = 0.696, $p = 0.003$) were confirmed as significant indicator taxa for the broader Group A+B (Figure 4, highlighted in green).

In contrast, another cluster of genera exhibited significant positive correlations with these nutrient and organic condition parameters, indicating them as highly pollution-tolerant. Genera such as *Chironomus*, *Cloeon*, *Dicrotendipes*, and *Glyptotendipes* were classified as same cluster. *Glyptotendipes* (Indicator Value = 0.761, $p = 0.001$) was confirmed as a definitive indicator for Group C, showing a strong affinity for elevated organic loading (Figure 4, highlighted in red). Furthermore, *Cloeon* (Indicator Value = 0.687, $p = 0.019$), *Dicrotendipes* (Indicator Value = 0.815, $p = 0.003$) and *Chironomus* (Indicator Value = 0.798, $p = 0.006$) were identified as significant indicators for the combined Group B+C (Figure 4, highlighted in orange).

4. Discussion

In this study, aquatic insect communities were initially analyzed across 60 samples using *COI* metabarcoding. Among these, 12 samples had insufficient sequencing depth ($< 3,000$ aquatic insect reads) and were therefore excluded based on rarefaction analysis, as they did not adequately represent the assemblages (see Table 1). This limited recovery in certain samples is primarily attributed to biological characteristics of benthic macroinvertebrates—specifically their highly developed chitinous exoskeletons—which can lead to stochastic DNA detection from the water column, sediment, or gravel surfaces depending on the timing of molting or local biomass levels. Unlike vertebrates that continuously shed DNA through mucus or skin, the chitinous exoskeletons of many invertebrates act as a physical barrier that limits the release of DNA into the water column (Trimbos et al. 2021; Gleason et al. 2021; Poyntz-Wright et al. 2024). This structural characteristic often results in lower eDNA concentrations compared to other taxa, leading to the insufficient sequencing depth observed in certain samples. Despite the exclusion of these samples, the substantial sequencing depth achieved in the remaining 48 samples—totaling over 7.2 million high-quality reads—offers more than enough statistical power to clearly elucidate the impact of water quality gradients on community structures of aquatic insects. However, as this study relied solely on eDNA data, it may not fully capture organismal abundance or life-stage variability. Integrating eDNA approaches with conventional field surveys will improve ecological resolution and enable more robust, quantitative biomonitoring frameworks (Bush et al., 2019; Poyntz-Wright et al., 2024). Yet, previous studies on macroinvertebrates suggest that eDNA performs as well as or even outperforms conventional methods in riverine environmental assessment (Fernández et al., 2019; Uchida et al., 2020).

The observed alpha diversity patterns provide a nuanced understanding of how benthic macroinvertebrate communities respond to water quality degradation. Our results demonstrate a significant decline in species richness from pristine sites (Group A) to impacted sites (Groups B and C). In contrast, other alpha diversity indices exhibit relative stability across all water quality groups. This discrepancy suggests that the community response in this system is primarily driven by the selective loss of sensitive taxa rather than a fundamental reorganization of community dominance patterns. Elevated concentrations of TP, TOC, and NH₃-N are known to alter habitat conditions through eutrophication, increased microbial respiration, and subsequent oxygen depletion (Dodds and Smith, 2016; Mosley et al., 2023). Such changes impose strong physiological constraints on taxa with narrow tolerance ranges, particularly those requiring well-oxygenated and oligotrophic conditions. As a result, these taxa are progressively excluded from degraded environments, leading to a diminished species richness without destabilizing the proportional abundance of the remaining tolerant community. We should note that the factor of season (May vs. September) explained 21.5% of the variation in community structure ($R^2 = 0.215$), which was lower than the explanatory power of water quality conditions ($R^2 = 0.272$). This indicates that, although seasonal variation influences community differences, environmental pressures associated with water quality degradation exert a stronger and more consistent effect on aquatic insect community structure.

The superior explanatory power of Weighted UniFrac ($R^2 = 56.3\%$) over Bray-Curtis ($R^2 = 36.6\%$) suggests that accounting for evolutionary relationships provides a more robust framework for understanding community assembly across distinct environmental conditions (Lozupone and Knight, 2005). In this study, Bray-Curtis distances encountered a discriminatory limit in disturbed environments, failing to statistically distinguish between Group B and Group C ($p > 0.05$). This indicates that while these impacted sites may appear taxonomically similar—likely due to the dominance of a shared suite of tolerant taxa—their

phylogenetic structures remain responsive to further environmental shifts (Ge et al., 2022; Serrana et al., 2025). Furthermore, the specific sensitivity of Weighted UniFrac to nitrogen-related stressors (NH₃-N and TN) indicates that the influence of these variables is more clearly reflected when considering the phylogenetic relationships within the community. This suggests that nitrogen sensitivity in aquatic insects may be a phylogenetically conserved trait, consistent with previous findings that physiological responses to environmental stressors in benthic macroinvertebrates are strongly driven by phylogenetic parameters rather than random taxonomic variation (Buchwalter et al., 2008).

Taxonomic composition shifted along the water quality gradient, reflecting distinct community structures. Our results demonstrate a dramatic transition in dominance, where the balanced assemblage of Ephemeroptera and Diptera in Group A progressively shifts to a nearly monolithic dominance of Diptera (exceeding 80%) in Group C. This pattern is consistent with well-established biomonitoring approaches, in which sensitive taxa such as Ephemeroptera, Plecoptera, and Trichoptera (EPT taxa) decline under degraded conditions, while tolerant groups—particularly Chironomidae within Diptera—proliferate in response to organic enrichment and reduced habitat quality (Barasa et al., 2025; Rosenberg and Resh, 1992). The near disappearance of Plecoptera and the drastic reduction of Trichoptera and Ephemeroptera in Group C further support the sensitivity of these taxa to elevated nutrient and organic loads.

Our findings are broadly consistent with previous studies demonstrating the efficacy of aquatic insect and macroinvertebrate assemblages for water quality assessment. Fernández et al. (2019) reported a significant positive correlation between eDNA-based and morphology-based diversity indices, validating the reliability of eDNA metabarcoding for riverine bioassessment. Similarly, Uchida et al. (2020) demonstrated that eDNA metabarcoding not only detected a higher taxonomic diversity than traditional Surber net sampling but also that eDNA-derived EPT and Chironomidae indices were closely associated with total nitrogen

levels. These observations are consistent with our findings that aquatic insect communities—characterized by *COI* metabarcoding—responded clearly to nutrient and organic pollution gradients, particularly TP, TOC, NH₃-N, and TN, evidenced by the attrition of EPT taxa and increased dominance of Diptera in degraded sites. Furthermore, Rodríguez et al. (2024) confirmed that aquatic insect assemblages effectively reflect physicochemical degradation, including phosphorus-related water quality changes, reinforcing the utility of these taxa as robust bioindicators of freshwater ecosystem health.

The identification of specific indicator taxa provides further evidence of the nuanced ecological responses within the community. Environmentally sensitive taxa, including *Ecdyonurus*, *Ephemera*, and *Teloganopsis*, showed strong negative correlations with pollution variables, highlighting their role as core indicators of high ecological integrity. Conversely, the proliferation of tolerant genera like *Chironomus*, *Dicrotendipes*, and *Glyptotendipes* in Group C emphasizes their physiological resilience to organically enriched and hypoxic conditions. Notably, the detection of *Rheopelopia* (Family Chironomidae) as an indicator for Group A challenges the general perception of Chironomids as universally pollution-tolerant taxa. Previous studies have reported that *Rheopelopia* species typically inhabit coarse substrates, such as large cobbles in swift river currents, reflecting a preference for physically stable and well-oxygenated environments (Klink, 1982). By capturing such ecological nuances within broadly tolerant groups, our study demonstrates that substrate-derived eDNA assessment can offer precise ecological signals, enabling a more granular evaluation of freshwater ecosystem health.

5. Conclusion

456 In this study, we demonstrated that benthic substrate-associated eDNA metabarcoding of
457 aquatic insect communities serves as a powerful and reliable tool for representing water quality
458 gradients in freshwater ecosystems. The significant decline in species richness in response to
459 increasing pollution loads (TP, TOC, and NH₃-N) indicates that environmental stress acts as a
460 primary driver for the selective loss of sensitive taxa, leading to an initial depletion of biological
461 diversity. Our results further suggest that sensitivity to nitrogen-related stressors is a conserved
462 trait within specific evolutionary lineages, which is more effectively captured by phylogenetic
463 distance metrics than by traditional taxonomic counts. Furthermore, the identification of high-
464 resolution indicator taxa, such as the pristine water associated *Rheopelopia*, highlights the
465 superior diagnostic capability of eDNA metabarcoding over traditional morphological
466 assessments. Ultimately, these results provide a critical scientific foundation for the transition
467 toward next-generation biomonitoring frameworks, offering a powerful, data-driven tool for
468 the management and restoration of riverine biodiversity.

469 **CRedit authorship contribution statement**

470 **HK:** Conceptualization, data curation, formal analysis, funding acquisition, investigation,
471 methodology, project administration, visualization, writing – original draft. **H-kC:**
472 Conceptualization, data curation, formal analysis, investigation, methodology, project
473 administration, visualization, writing – original draft. **ML:** Data curation, formal analysis,
474 investigation, methodology, project administration, writing – review and editing. **YRK:**
475 Formal analysis, investigation, methodology, writing – review and editing. **SYH:** Formal
476 analysis, investigation, methodology, writing – review and editing. **JHK:** Conceptualization,
477 writing – review and editing. **KSC:** Writing – review and editing. **K-RK:** Writing – review
478 and editing. **HJL:** Conceptualization; data curation, funding acquisition, methodology, project
479 administration, supervision, writing – review and editing. We agree with publication of the
480 final version of this manuscript.

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484 **Data Archiving Statement**

485 The raw sequencing data underlying the main results of this study will be archived in the
486 Zenodo Repository and made publicly available upon acceptance of the manuscript
487 (<https://doi.org/10.5281/zenodo.22025702>).

488 **Appendix A. Supporting Information**

489 Supporting Information for this article can be found online at <https://doi.org/XXXX/XXXX>.

490 **Declaration of competing interest**

491 The authors have declared that no competing interest exist.

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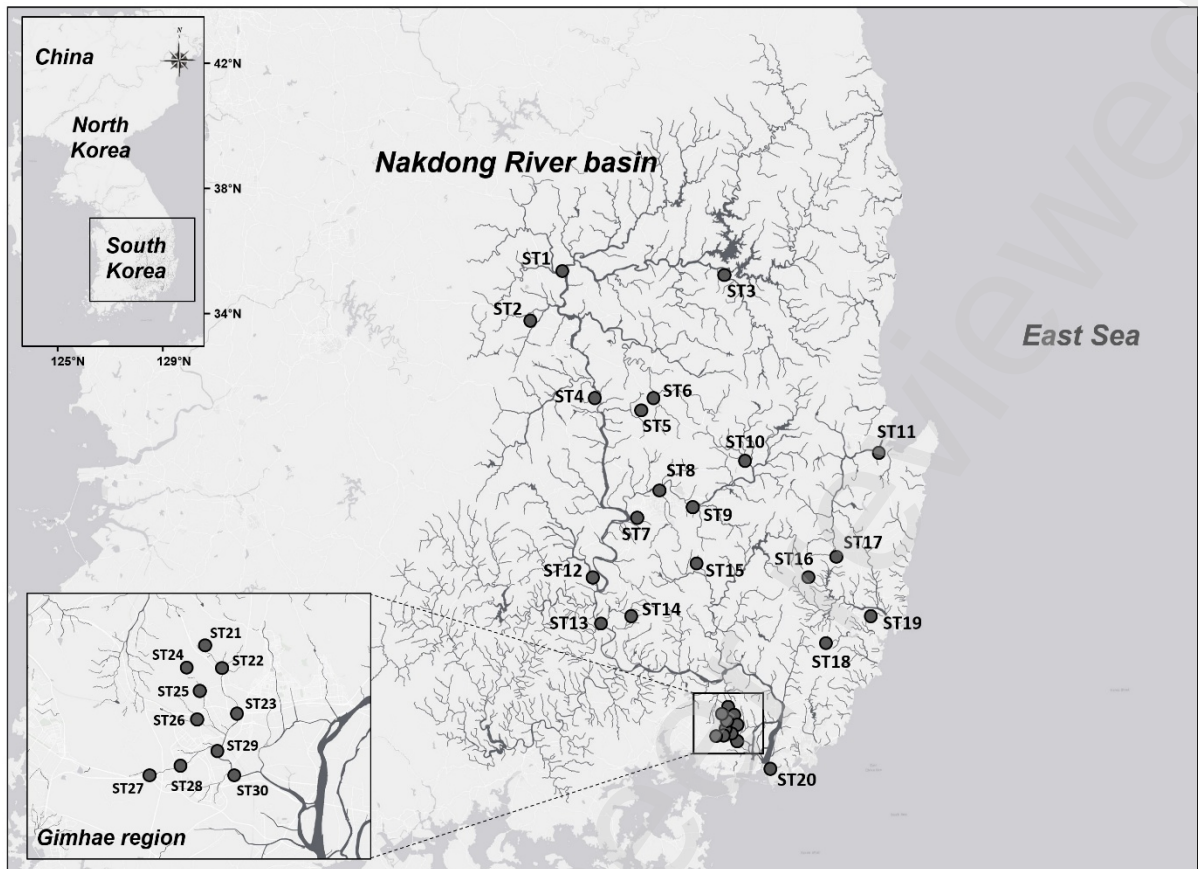
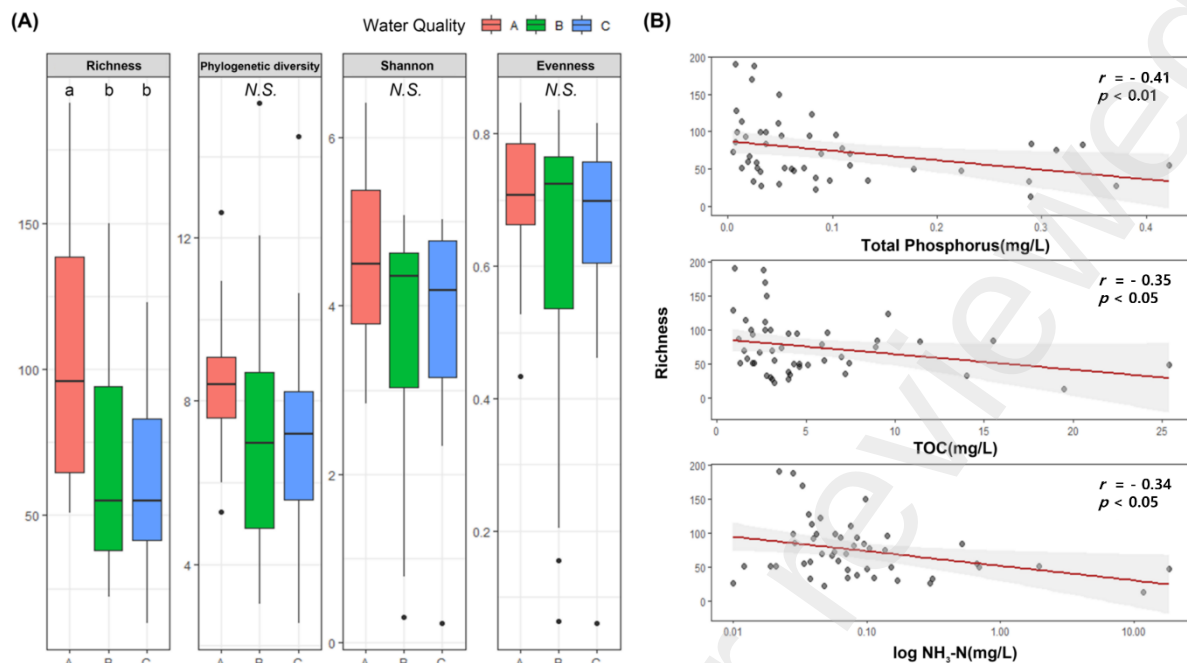
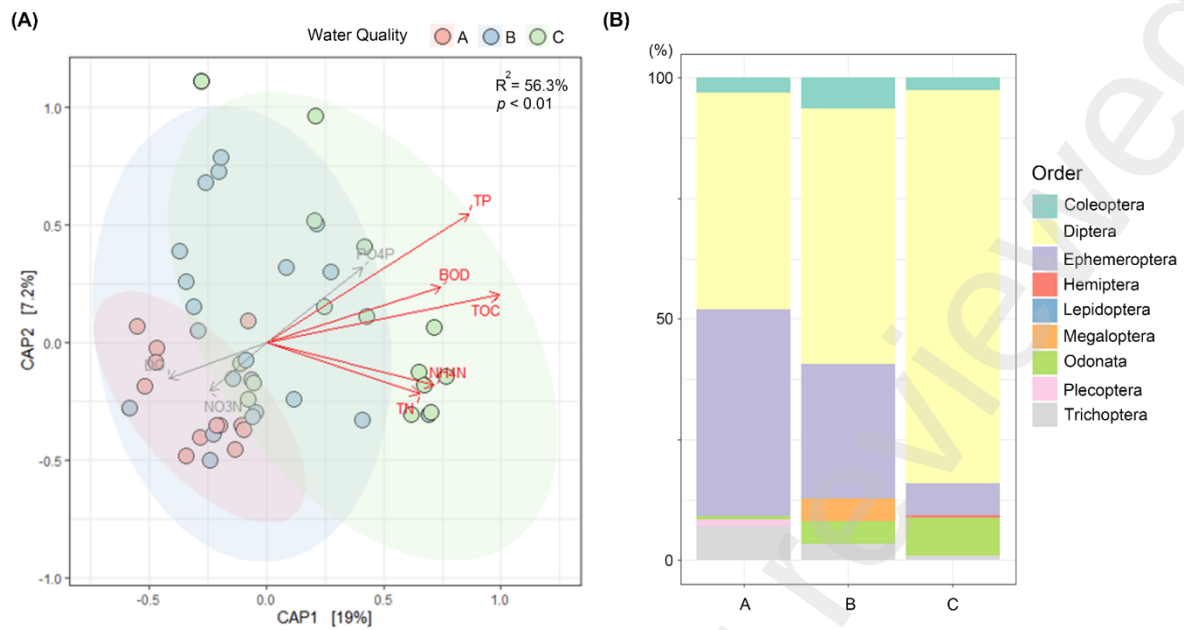


Figure 1. Geographic distribution of sampling sites in the Nakdong River basin, South Korea. The study included 30 sampling sites (ST1–ST30) distributed across the main stream and tributaries of the Nakdong River.



649

650 **Figure 2. Comparison of alpha diversity indices and their correlations with key**
 651 **environmental variables.** (A) Boxplots of alpha diversity indices across the three water
 652 quality groups (A, B, and C). Different lowercase letters (a, b) above the boxes indicate
 653 statistically significant differences between groups (Kruskal-Wallis test followed by post-hoc
 654 analysis, $p < 0.05$), while N.S. denotes no significant difference. (B) Linear regression analysis
 655 between aquatic insect richness and major physicochemical parameters: TP, TOC, and log-
 656 transformed NH₃-N). The shaded areas represent the 95% confidence intervals.



657

658 **Figure 3. Community structure and taxonomic composition of aquatic insects across**
 659 **water quality groups.** (A) CAP analysis based on Weighted UniFrac distances, illustrating the
 660 shifts in abundance-weighted phylogenetic structure. The red arrows represent significant
 661 environmental variables. (B) Relative abundance of aquatic insect orders across the three water
 662 quality groups.

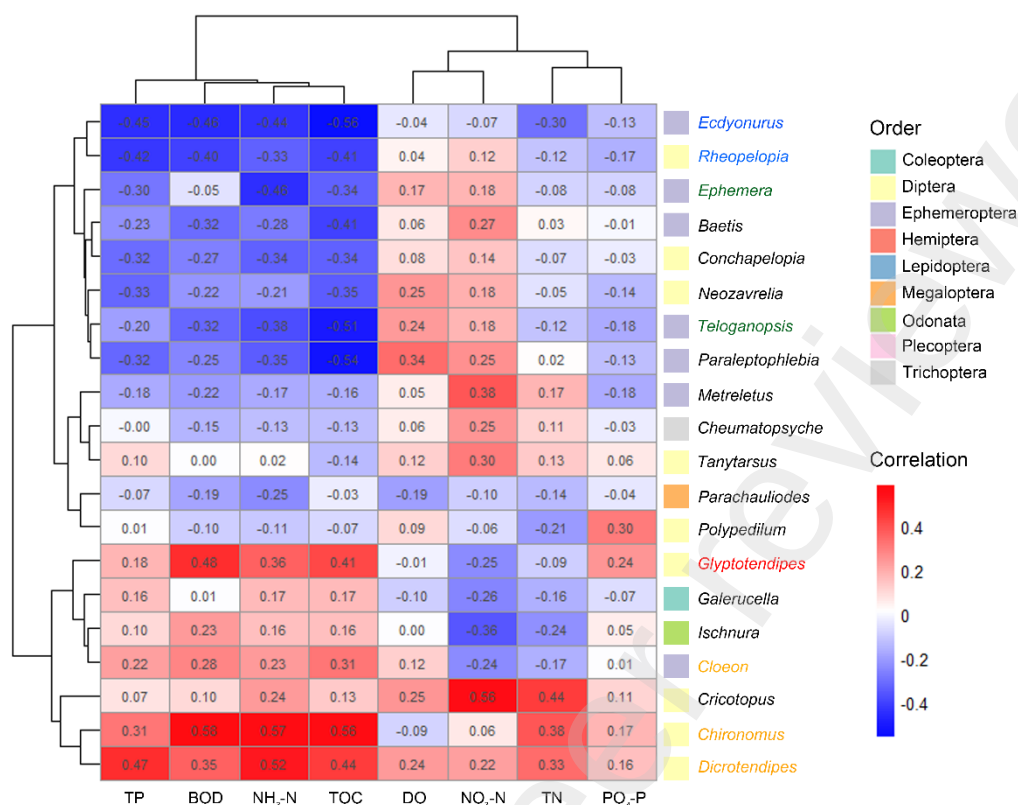


Figure 4. Correlation analysis between dominant aquatic insect genera and water quality parameters. The heatmap illustrates the Spearman correlation coefficients between the relative abundance of major genera and physicochemical variables. Taxa on the y-axis are color-coded by their respective orders. Both taxa and environmental variables are organized by hierarchical clustering based on Euclidean distances. The color scale represents the strength of the correlation, with red indicating a positive correlation (pollution-tolerant taxa) and blue indicating a negative correlation (pollution-sensitive taxa).

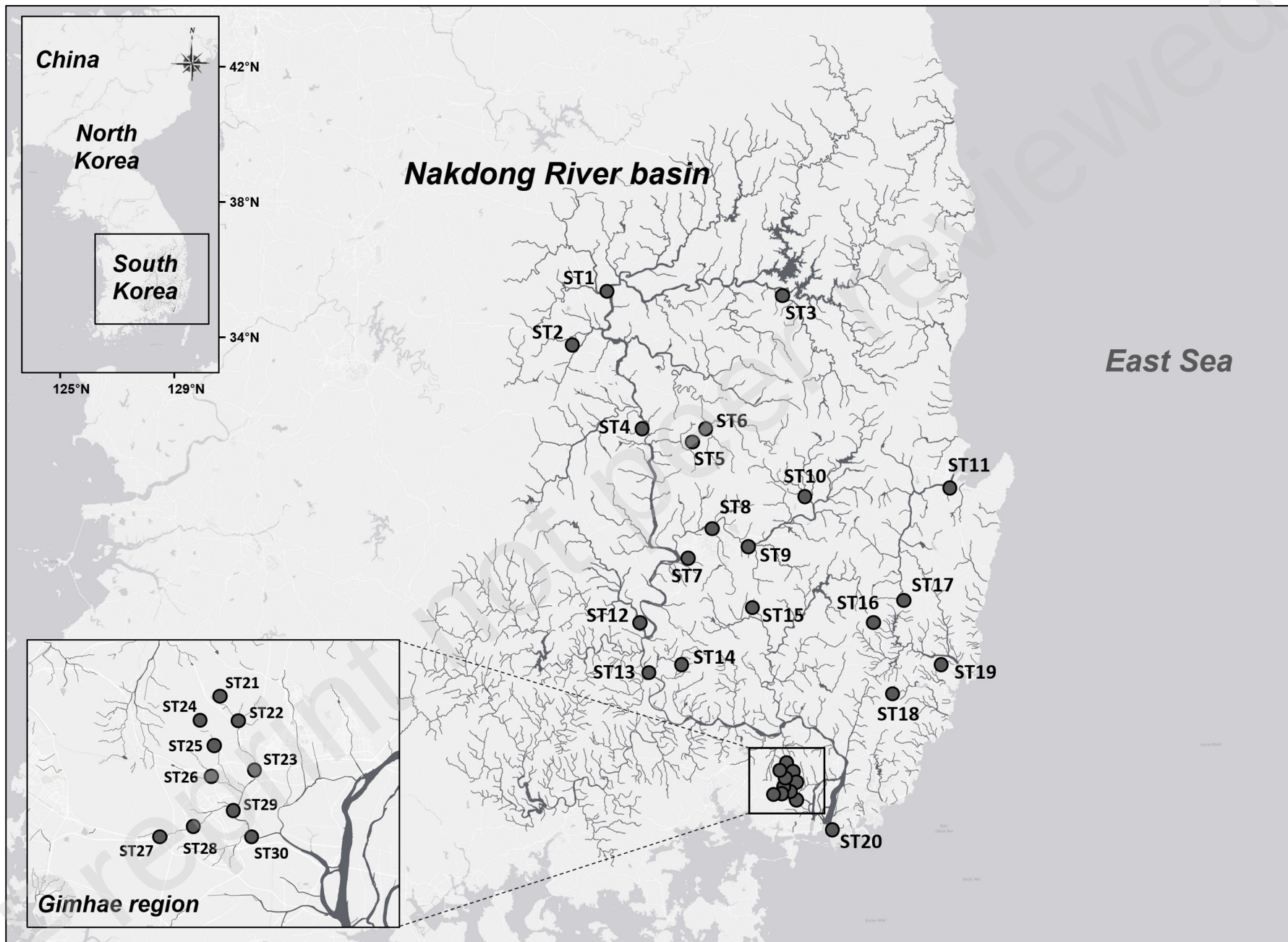
671 **Table 1.** Detailed information on the sampling sites, including geographical coordinates, sample ID (S: sampled in spring [May]; A: sampled
672 in autumn [September]), and whether the sample was included in final metabarcoding analysis. The 12 samples were omitted from the
673 metabarcoding analysis because they had insufficient sequencing depth (< 3,000 aquatic insect reads) based on rarefaction analysis.

Sampling site	Geographical coordinate	Sample ID (Spring)	Aquatic insect reads	Included in analysis	Sample ID (Autumn)	Aquatic insect reads	Included in analysis
ST1	N36°32'16.01", E128°15'23.67"	S E 1 S COI	24,650	Yes	A E 1 S COI	6,587	Yes
ST2	N36°23'40.08", E128°08'28.67"	S E 2 S COI	2,937	No	A E 2 S COI	3,866	Yes
ST3	N36°31'08.69", E128°50'19.24"	S E 3 S COI	24,062	Yes	A E 3 S COI	32,212	Yes
ST4	N36°10'06.09", E128°22'11.57"	S E 4 S COI	913	No	A E 4 S COI	7,821	Yes
ST5	N36°08'28.24", E128°32'00.85"	S E 5 S COI	6,879	Yes	A E 5 S COI	1,157	No
ST6	N36°10'00.33", E128°34'09.32"	S E 6 S COI	6,218	Yes	A E 6 S COI	6,255	Yes
ST7	N35°49'30.00", E128°30'39.59"	S E 7 S COI	10,995	Yes	A E 7 S COI	142	No
ST8	N35°54'10.51", E128°35'34.62"	S E 8 S COI	6,357	Yes	A E 8 S COI	1,833	No
ST9	N35°51'06.97", E128°42'27.96"	S E 9 S COI	6,772	Yes	A E 9 S COI	894	No
ST10	N35°58'51.00", E128°54'01.58"	S E 10 S COI	27,292	Yes	A E 10 S COI	1,684	No
ST11	N35°59'56.36", E129°22'11.33"	S E 11 S COI	32,796	Yes	A E 11 S COI	12,282	Yes
ST12	N35°39'03.04", E128°20'53.94"	S E 12 S COI	18,931	Yes	A E 12 S COI	8,150	Yes
ST13	N35°31'09.86", E128°22'35.74"	S E 13 S COI	23,185	Yes	A E 13 S COI	7,179	Yes
ST14	N35°32'16.65", E128°28'50.41"	S E 14 S COI	17,756	Yes	A E 14 S COI	3,049	Yes
ST15	N35°41'05.48", E128°43'16.00"	S E 15 S COI	6,603	Yes	A E 15 S COI	3,977	Yes
ST16	N35°38'27.15", E129°06'49.77"	S E 16 S COI	9,969	Yes	A E 16 S COI	541	No
ST17	N35°41'41.75", E129°12'19.82"	S E 17 S COI	4,115	Yes	A E 17 S COI	2,401	No
ST18	N35°26'58.80", E129°10'13.27"	S E 18 S COI	7,474	Yes	A E 18 S COI	15,031	Yes
ST19	N35°31'59.72", E129°20'03.22"	S E 19 S COI	5,022	Yes	A E 19 S COI	45	No
ST20	N35°05'01.30", E128°57'58.05"	S E 20 S COI	1,351	No	A E 20 S COI	20,301	Yes
ST21	N35°15'40.47", E128°49'06.36"	S E 21 S COI	18,772	Yes	A E 21 S COI	5,470	Yes
ST22	N35°14'52.74", E128°49'51.94"	S E 22 S COI	3,518	Yes	A E 22 S COI	10,983	Yes
ST23	N35°13'09.08", E128°50'28.42"	S E 23 S COI	15,428	Yes	A E 23 S COI	31,750	Yes
ST24	N35°14'58.34", E128°48'21.17"	S E 24 S COI	14,248	Yes	A E 24 S COI	40,581	Yes
ST25	N35°14'06.11", E128°48'56.70"	S E 25 S COI	23,944	Yes	A E 25 S COI	16,471	Yes
ST26	N35°13'04.96", E128°48'42.84"	S E 27 S COI	11,832	Yes	A E 26 S COI	5,212	Yes
ST27	N35°11'04.68", E128°46'33.65"	S E 26 S COI	18,296	Yes	A E 27 S COI	17,382	Yes
ST28	N35°11'19.08", E128°47'58.66"	S E 28 S COI	30,249	Yes	A E 28 S COI	16,587	Yes
ST29	N35°11'52.34", E128°49'34.31"	S E 29 S COI	30,044	Yes	A E 29 S COI	1,177	No
ST30	N35°10'58.28", E128°50'19.99"	S E 30 S COI	9,643	Yes	A E 30 S COI	8,990	Yes

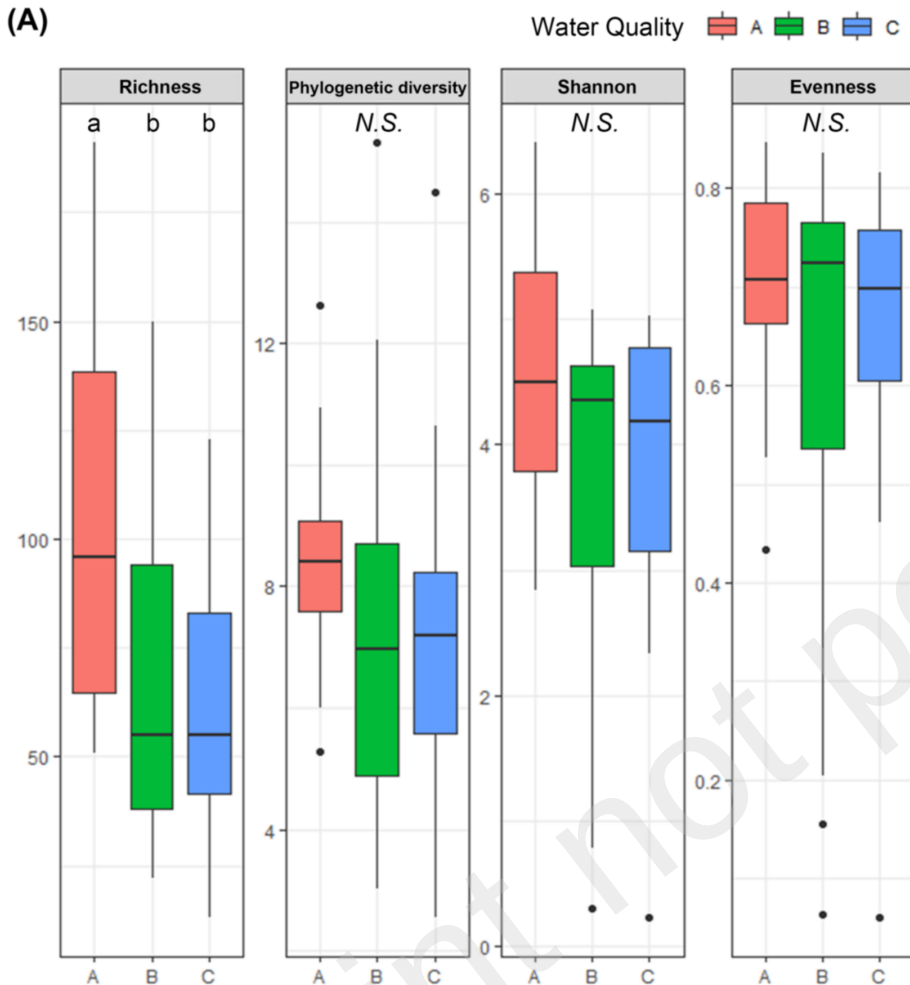
675 **Table 2.** Physicochemical parameters across the water quality groups.

Parameter	A (n=12)	B (n=21)	C (n=15)
BOD (mg/L)	0.867 ± 0.166	2.195 ± 0.236	9.180 ± 1.508
DO (mg/L)	9.319 ± 0.529	10.338 ± 0.954	8.139 ± 0.935
NH ₃ -N (mg/L)	0.033 ± 0.003	0.071 ± 0.008	2.355 ± 1.381
NO ₃ -N (mg/L)	0.809 ± 0.479	1.726 ± 0.479	1.176 ± 0.347
TN (mg/L)	0.911 ± 0.171	2.017 ± 0.476	5.010 ± 1.699
TP (mg/L)	0.018 ± 0.002	0.063 ± 0.007	0.208 ± 0.034
PO ₄ -P (mg/L)	0.000 ± 0.000	0.000 ± 0.000	0.040 ± 0.020
TOC (mg/L)	1.850 ± 0.191	3.462 ± 0.239	10.360 ± 1.541

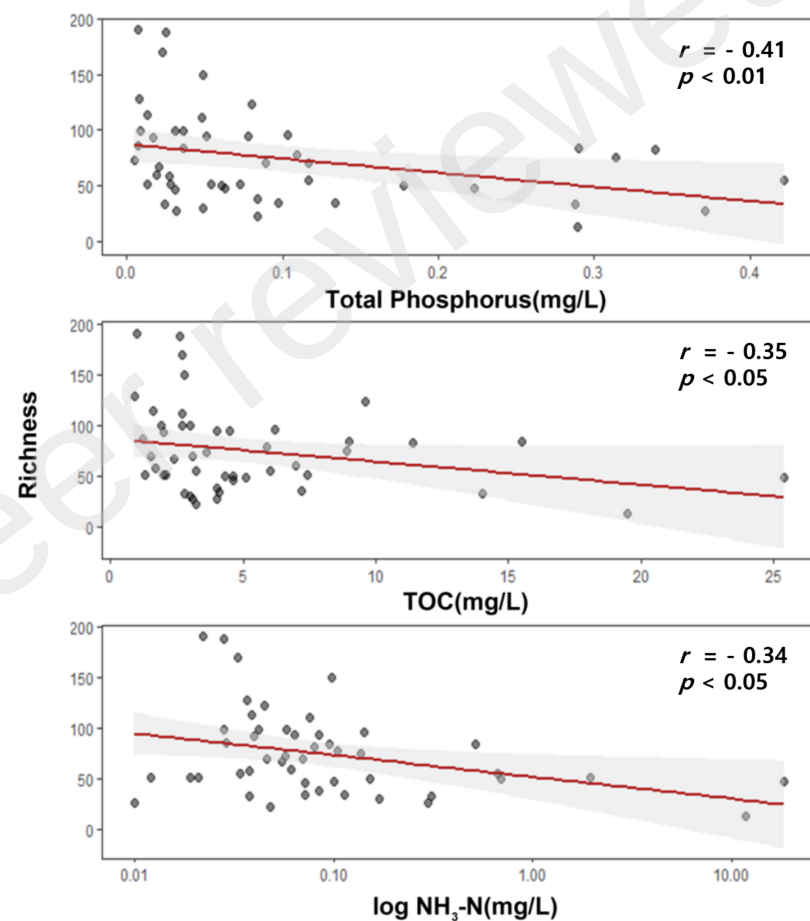
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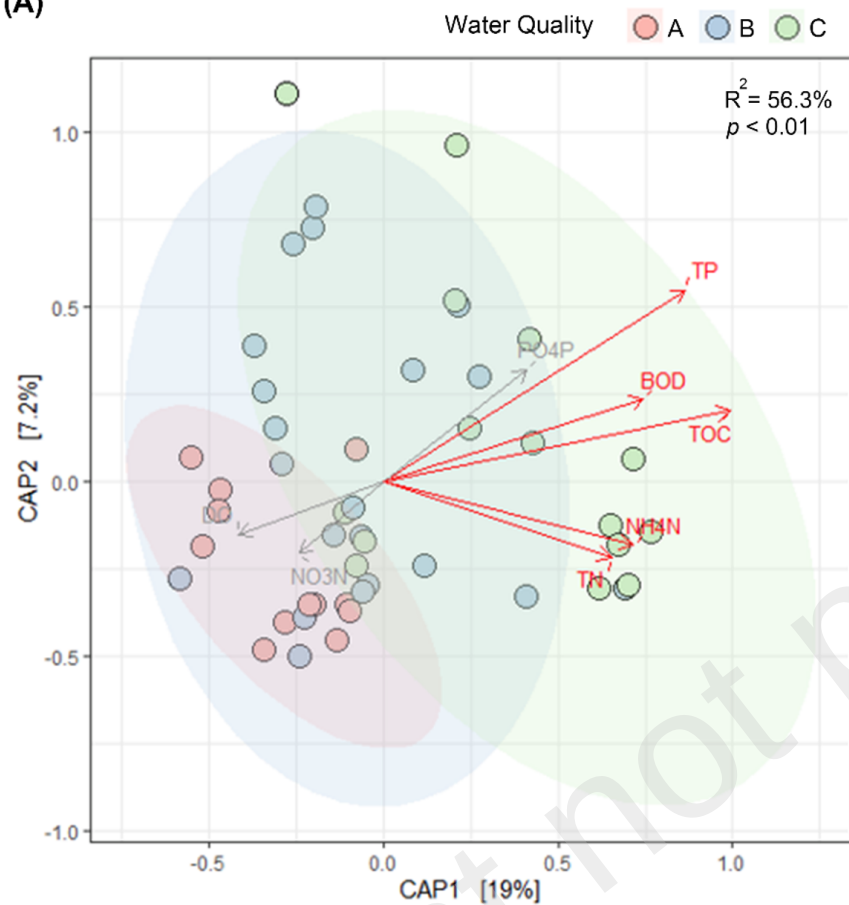
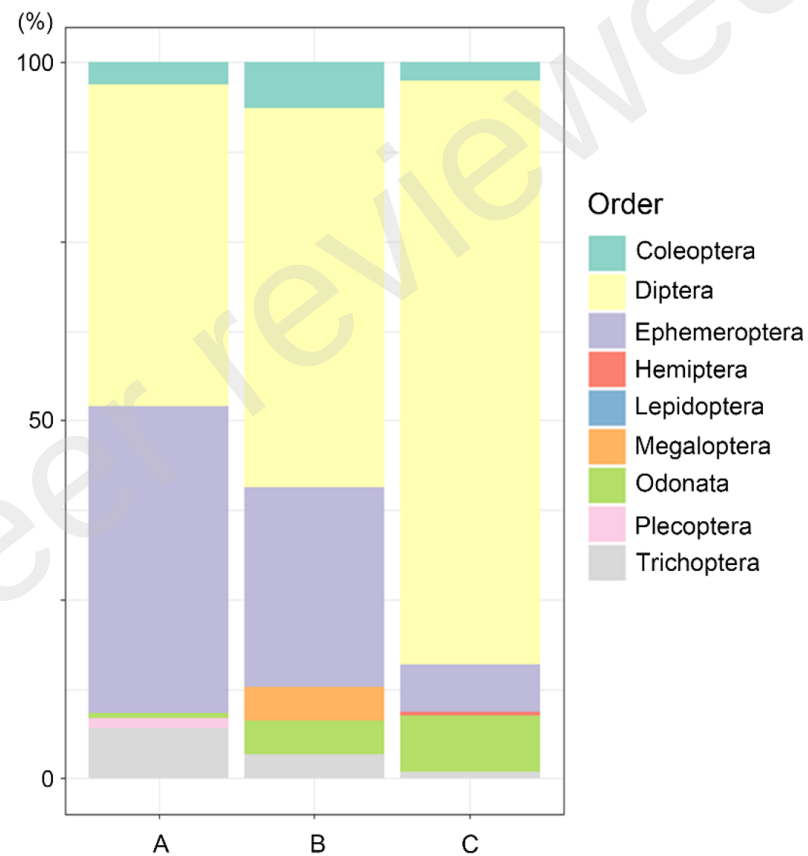


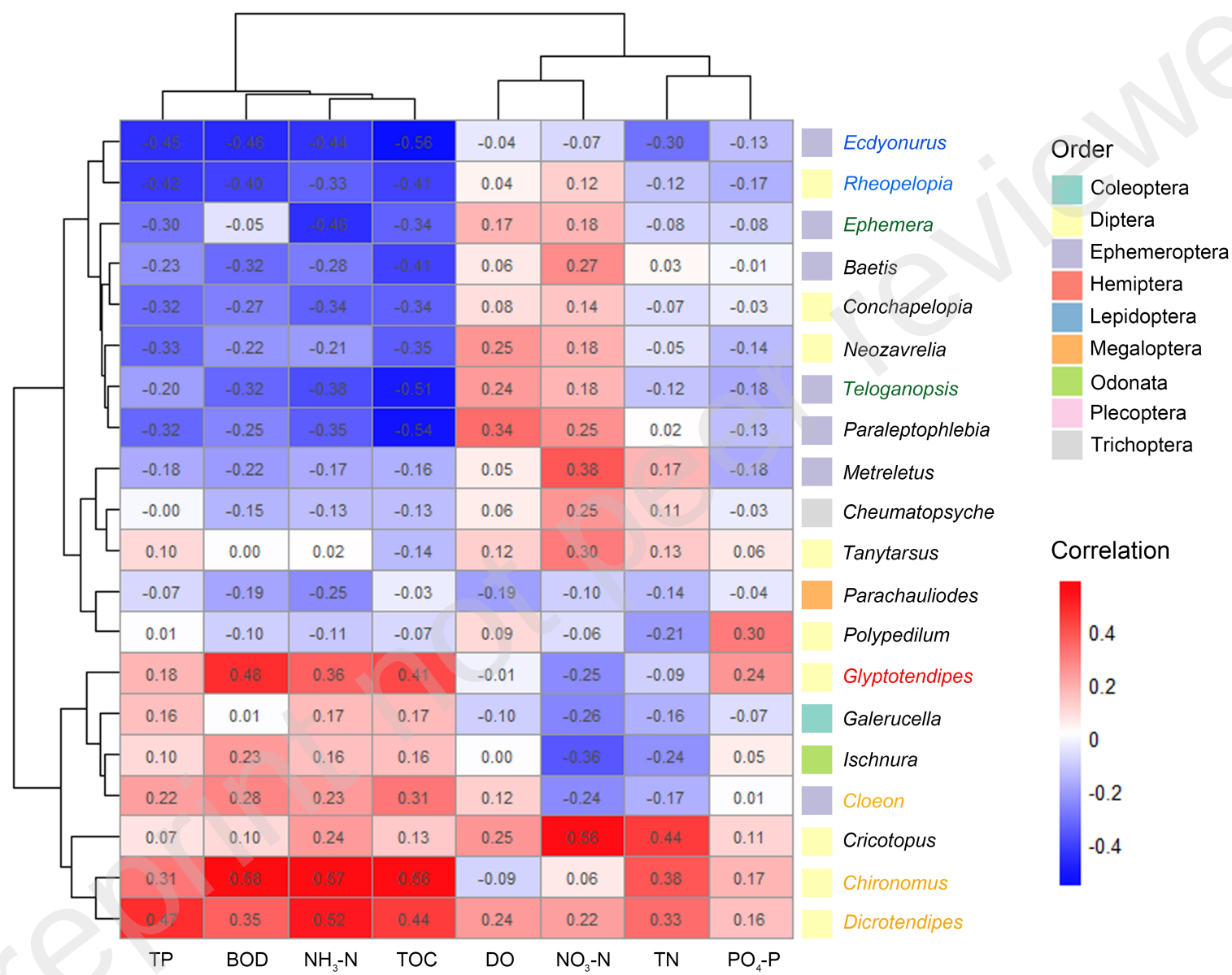
(A)



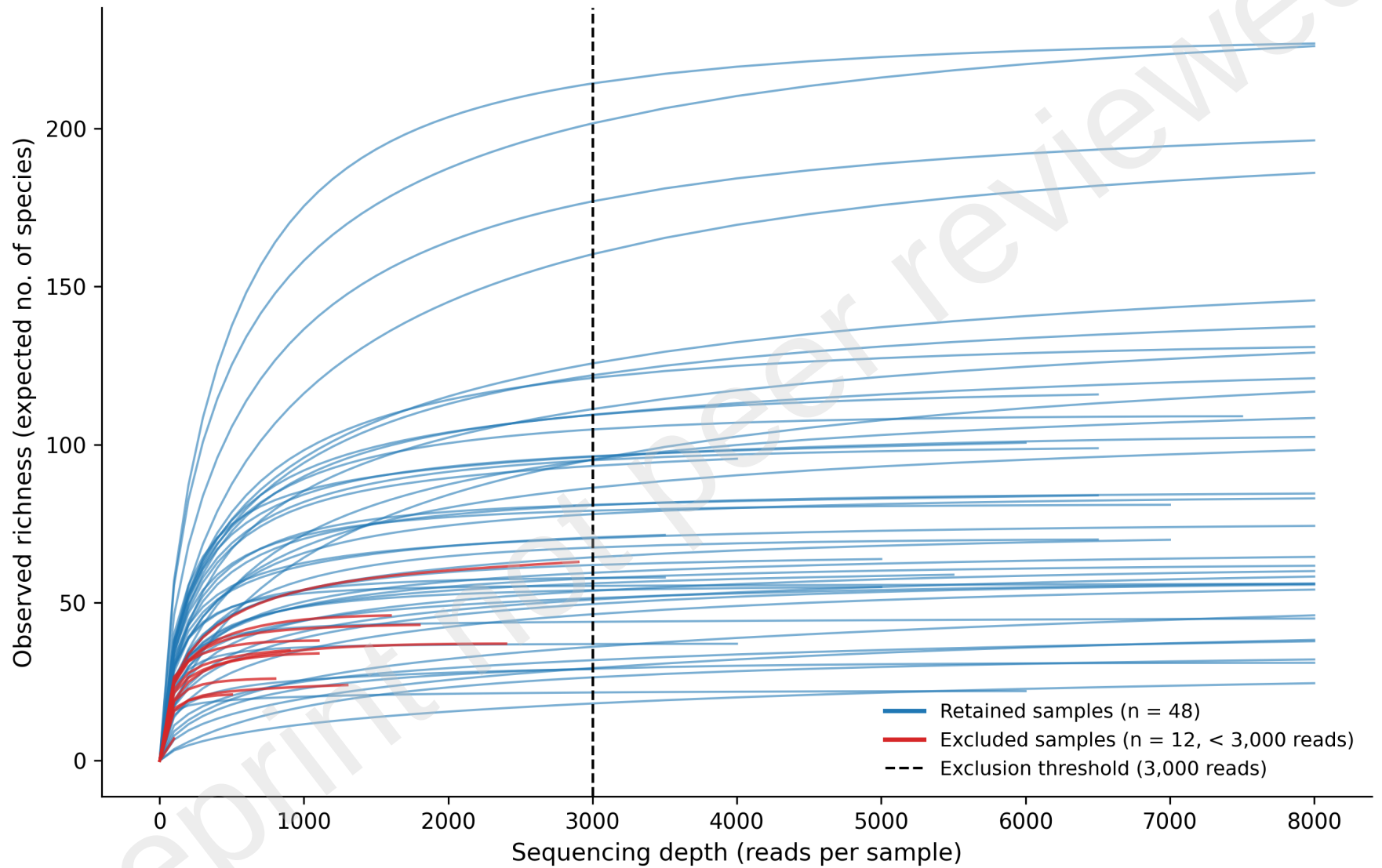
(B)



(A)**(B)**



Alpha rarefaction curves (0-8,000 reads, zoomed near threshold)



Water Quality

A

B

C

